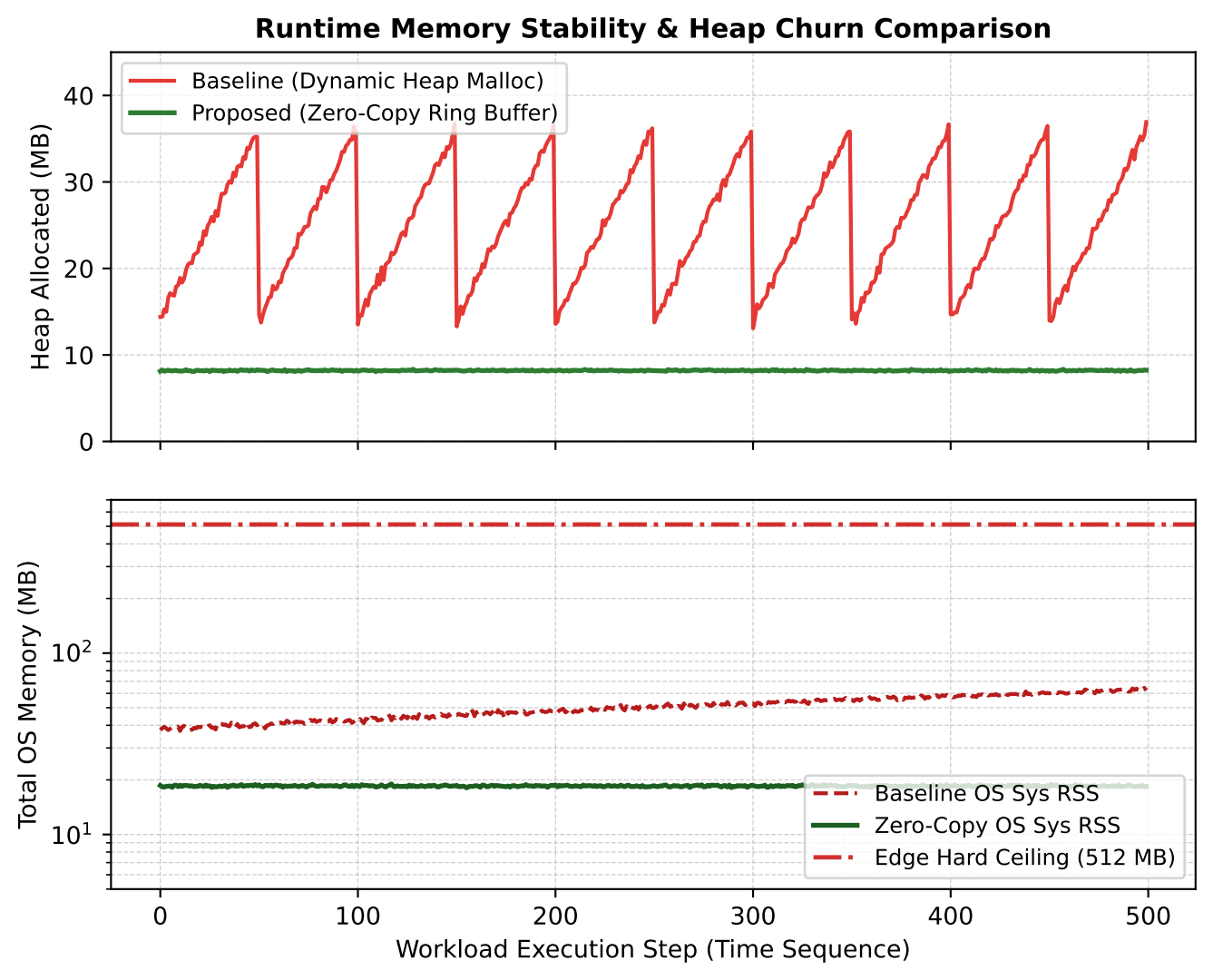
Runtime Memory Stability & Heap Churn Comparison
Baseline (Dynamic Heap Malloc)
Proposed (Zero-Copy Ring Buffer)
Heap Allocated (MB)
0
10
20
30
40
Total OS Memory (MB)
10^2
10^1
Baseline OS Sys RSS
Zero-Copy OS Sys RSS
Edge Hard Ceiling (512 MB)
0
100
200
300
400
500
Workload Execution Step (Time Sequence)